

Acknowledgments

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Khoa Nguyen Taoyuan, April 2026

摘要

在新領域或特定領域中，命名實體識別（Named Entity Recognition, NER）通常需要人工定義實體類型並標註資料，此過程不僅耗費成本，且需要領域專家的參與。這項限制促使了開放世界命名實體識別（**Open-world NER**）的發展，其目標不僅是從未標註的文本中偵測實體提及（entity mentions），更要自動發掘並命名潛在的實體類型。

在本論文中，我們探討在領域轉移（domain shift）下的非監督式開放世界 NER。在此情境下，實體類型並未預先定義，而系統效能則取決於高召回率（high-recall）的提及偵測，以及具鑑別力的實體表徵（entity representations）以進行分群。基於 OWNER 框架，我們提出三項改進。首先，我們以多視角分治轉移（multi-view divide-and-transfer）框架 [1] 取代 OWNER 的提及偵測機制：一個共享的文本編碼器（text encoder）連接三個互補的解碼頭（decoding heads）—— BIO 標註、起始/結束邊界標註，以及連繫/斷開（tie/break）跨度連結。將這三者的結果進行集成（ensembled）與過濾，以產生穩健且高召回率的實體跨度（entity spans）。其次，在實體分類（entity typing）方面，我們引入了軟提示實體編碼（soft-prompt entity encoding），透過在 BERT 風格的編碼器前附加可訓練的虛擬詞元（virtual tokens），並利用提示詞中 [MASK] 位置的上下文隱藏狀態（contextualized hidden state）來表示每個實體提及，從而改善開放世界分群的任務條件調節（task conditioning）。第三，我們提出了一個分群命名框架，為給定的語料庫生成人類可讀的實體類型名稱：我們透過最大邊際相關性（maximal marginal relevance, MMR）挑選出具多樣性的分群範例（cluster exemplars），接著使用遮蔽語言模型（masked language modeling, MLM）共識或本地端大型語

言模型（Ollama）推論出簡潔的類別名稱，並以 BERTScore 評估命名品質。

結合上述元件，我們構建了一個端到端（end-to-end）、更具穩健性與可解釋性的非監督式開放世界 NER 系統。該系統不僅能對實體進行分群，還能夠為新的語料庫產生具可讀性的類型清單（type inventories）。

關鍵字：開放世界命名實體識別（Open-world Named Entity Recognition）；非監督式 NER（Unsupervised NER）；領域轉移（Domain Shift）；提及偵測（Mention Detection）；多視角解碼（Multi-view Decoding）；實體分類（Entity Typing）；軟提示（Soft Prompting）；表徵學習（Representation Learning）；分群（Clustering）；分群命名（Cluster Naming）；遮蔽語言模型（Masked Language Modeling）；本地大型語言模型（Local LLM）

Abstract

Named Entity Recognition (NER) in new or specialized domains often requires manually defining entity types and annotating data, a process that is costly and demands domain expertise. This limitation motivates **open-world NER**, where the goal is not only to detect entity mentions from unlabeled text but also to **discover and name latent entity types** automatically. In this thesis, we study unsupervised open-world NER under domain shift, where entity types are not predefined and performance depends on both high-recall mention detection and discriminative entity representations for clustering. Building on the OWNER framework, we propose three enhancements. First, we replace OWNER’s mention detection with a multi-view divide-and-transfer framework [1]: a shared text encoder feeds three complementary decoding heads—BIO tagging, start/end boundary tagging, and tie/break span linkage—which are ensembled and filtered to produce robust, high-recall entity spans. Second, for entity typing, we introduce soft-prompt entity encoding by prepending trainable virtual tokens to a BERT-style encoder and representing each mention via the contextualized hidden state at the prompt’s [MASK] position, improving task conditioning for open-world clustering. Third, we propose a cluster naming framework to generate human-readable entity type names for a given corpus: we select diverse cluster exemplars via maximal marginal relevance (MMR) and infer concise category names using either masked language modeling (MLM) consensus or a local LLM (Ollama), and evaluate naming quality with BERTScore. Together, these components form an end-to-end, more robust and interpretable unsupervised

open-world NER system that not only clusters entities but also produces readable type inventories for new corpora.

Keywords: Open-world Named Entity Recognition; Unsupervised NER; Domain Shift; Mention Detection; Multi-view Decoding; Entity Typing; Soft Prompting; Representation Learning; Clustering; Cluster Naming; Masked Language Modeling; Local LLM



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Chapter 1

Introduction

Named Entity Recognition (NER) is a foundational task in natural language processing (NLP), underpinning a wide range of downstream applications including information extraction, knowledge graph construction, question answering, biomedical text mining, and document understanding. The objective of NER is to locate mentions of named entities in text and classify them into predefined semantic categories such as PERSON, ORGANIZATION, or LOCATION. Over the past decade, advances in deep learning and pretrained language models have pushed NER performance to near-human levels on standard benchmarks such as CoNLL-2003 [2] and OntoNotes [3].

Despite this progress, virtually all high-performing NER systems operate under a *closed-world assumption*: the set of entity types is fixed and known in advance, and models are trained on large amounts of manually annotated data conforming to that type inventory. This assumption, while convenient for benchmark evaluation, creates a fundamental barrier to deploying NER in practice. In real-world settings, new domains constantly emerge—scientific subfields, legal jurisdictions, industrial verticals, social media communities—each with its own vocabulary of relevant entities that may bear little resemblance to the coarse types defined in general-domain corpora. Annotat-

ing sufficient labeled data for every new domain is expensive, slow, and often requires specialized expert knowledge. Furthermore, even within a single domain, entity categories are not static: terminology evolves, new entity types emerge, and the granularity of the desired ontology varies by application.

These limitations motivate a shift toward *open-world NER*: a paradigm in which the system receives an unannotated corpus and is expected to (1) identify entity mention spans, (2) group those spans into coherent semantic categories without being told how many categories exist or what they are called, and (3) produce human-readable names for the discovered categories. The third requirement—*automatic entity type naming*—is often treated as an afterthought in existing work, yet it is critical for practical deployment. Clusters of entity mentions without interpretable labels cannot be validated by domain experts, integrated into knowledge bases, or used as supervision signals for downstream tasks. Naming is therefore not merely a cosmetic concern; it is what transforms unsupervised cluster assignments into a usable, inspectable entity type ontology.

This thesis presents **Span2Type**, an improved open-world NER pipeline built on top of the OWNER framework [4], with three targeted enhancements. Span2Type operates in three stages. First, the mention detection module from OWNER—which uses a single BIO sequence labeling head—is replaced with the Entity Detection module from the Divide-and-Transfer (DTrans) paradigm [1]. The DTrans module uses BERT with three complementary tagging schemes applied simultaneously: standard BIO tagging for global span context, Start-End (SE) binary classifiers for explicit boundary

detection, and Tie-Break (TB) classifiers that model whether adjacent tokens belong to the same span. Training all three views jointly with a Mean Teaching strategy (exponential moving average) improves domain transfer robustness, since entity span boundaries are more stable across domains than entity types. Second, the entity typing module encodes each detected span using the fill-in-the-blank template “*{sentence} {entity} is a [MASK].*” and extracts the [MASK]-position hidden state from a BERT-based encoder. This thesis introduces a soft prompt variant of this stage: a small number of trainable soft token embeddings are prepended to the template input while the base encoder is kept frozen, so that only the prompt parameters (not the full model) are updated during triplet margin training. The resulting embeddings are clustered with automatic k selection via the Bayesian Information Criterion (BIC). Third, the cluster naming module—which addresses the interpretability gap by generating human-readable type labels from cluster evidence—selects a diverse, representative set of exemplar sentences for each cluster using Maximal Marginal Relevance (MMR) and applies either a masked language model or a locally hosted large language model (LLM) to generate a concise, human-readable type name by aggregating predictions over the selected exemplars.

Span2Type requires no predefined type inventory and no target-domain annotations. The mention detector is trained on a source-domain corpus (CoNLL-2003 or Pile-NER [5]) and transferred to new domains without re-training. Experiments on CrossNER spanning five target domains demonstrate that replacing OWNER’s BIO-only mention detector with the DTrans

multi-view entity detector improves span recall across domains, and that MMR-based exemplar selection produces more specific and stable cluster names compared to OWNER’s original naming procedure.

The main contributions of this thesis are as follows:

- We integrate the DTrans multi-view Entity Detection module [1] into the OWNER open-world NER pipeline, replacing its single BIO mention detector with a joint BIO + Start-End + Tie-Break decoder that yields higher span recall in cross-domain settings.
- We propose an automatic cluster naming procedure that uses Maximal Marginal Relevance [6] to select a diverse set of exemplar sentences per cluster before querying an MLM or LLM, improving naming quality and stability over the centroid-nearest baseline used in the original OWNER system.
- We provide a systematic empirical evaluation on CrossNER across five target domains, comparing the improved pipeline against the original OWNER baseline and reporting entity detection F1, entity typing AMI, and cluster naming quality via BERTScore.

1.1 Motivation

The development of Span2Type is driven by three concrete challenges that arise when attempting to deploy NER systems beyond well-curated benchmark settings.

Challenge 1: Robust span detection across domains without target supervision. The first stage of any open-world NER pipeline is detecting entity mention spans in the target corpus without access to target-domain labels. The standard approach, used in OWNER, is to train a BIO sequence tagger on a labeled source domain (e.g., CoNLL-2003) and apply it directly to the target domain. While effective in closely related domains, single-view BIO taggers struggle when target domain text differs substantially in structure or vocabulary from the source. Entity boundary signals—such as whether a token starts a new mention or is connected to the preceding token—can be captured from multiple complementary perspectives. A multi-view detector that jointly trains BIO, Start-End, and Tie-Break decoders can exploit these different signals simultaneously, producing span candidates that are more robust to domain shift. This thesis addresses this challenge by replacing OWNER’s BIO-only mention detector with the DTrans multi-view Entity Detection module [1].

Challenge 2: Unknown and evolving entity type inventories. Even in domains where NER data exist, the type inventory may be incomplete or misaligned with the application’s needs. Standard NER benchmarks use small, coarse-grained label sets (typically four to eighteen types). Many real-world applications require finer granularity—distinguishing, for example, between DRUG, CHEMICAL, and DISEASE rather than a single MEDICAL type—or operate on entity categories that have no counterpart in existing ontologies, such as entities specific to proprietary product catalogs or newly coined technical terms. Purely supervised methods cannot discover new types; they can only

classify entities into the types they were trained on. Open-world NER, by contrast, makes no assumptions about the number or nature of entity types and can surface latent categories that would be missed by any closed-world system.

Challenge 3: The interpretability gap in unsupervised entity clustering.

Existing unsupervised entity typing methods typically produce cluster assignments without associated type names. The output of such a system is a mapping from entity spans to cluster identifiers (e.g., *Cluster 3*, *Cluster 7*), which carries no semantic content. Practitioners who wish to use these assignments must manually inspect cluster members, guess an appropriate label, and verify that the label is consistent across the cluster—a process that reintroduces the very human effort that unsupervised methods were supposed to eliminate. Automatic type naming bridges this interpretability gap: by generating a concise label such as PHARMACEUTICAL COMPANY or PROTEIN COMPLEX directly from cluster evidence, the system enables human experts to validate the discovered ontology and integrate it into downstream pipelines with minimal manual intervention.

Together, these three challenges motivate the three improvements this thesis makes to the OWNER pipeline. The DTrans multi-view entity detector addresses Challenge 1 by capturing complementary span boundary signals that transfer better across domains. The soft prompt entity typing variant improves entity embedding quality with drastically reduced parameter overhead, producing more type-discriminative representations for clus-

tering. The MMR-based cluster naming module addresses Challenge 3 by producing human-readable type names from diverse cluster evidence rather than leaving practitioners with anonymous cluster identifiers. Challenge 2—the unknown number of entity types—is handled by the BIC-guided k selection already present in OWNER, which this thesis retains unchanged.

1.2 Contributions

This thesis makes the following specific technical contributions:

- **Multi-view entity detection for open-world NER.** We replace the BIO-only mention detector in OWNER with the DTrans multi-view Entity Detection module [1], which jointly trains three complementary decoders—BIO, Start-End, and Tie-Break—on the source domain. The Mean Teaching strategy (exponential moving average) further stabilizes training, yielding more reliable span candidates when transferred to unseen target domains.
- **Two-stage prompt-initialized fine-tuning for entity typing.** We introduce a parameter-efficient variant of the entity typing stage that trains in two stages. Stage 1 prepends N_p trainable soft token embeddings to the fill-in-the-blank template and optimizes only these tokens at a high learning rate while the BERT backbone is strictly frozen, allowing rapid convergence without perturbing pre-trained weights. Stage 2 unfreezes the backbone at a drastically lower learning rate (1×10^{-5}) to prevent catastrophic forgetting, while the optimizer’s accumulated first- and second-moment statistics for the soft prompt are

preserved via `add_param_group()`. This two-stage schedule achieves the parameter efficiency of prompt-only tuning in early epochs and the full expressiveness of fine-tuning in later epochs.

- **MMR-based exemplar selection for automatic cluster naming.**

We propose selecting a diverse, centroid-relevant set of exemplar sentences per cluster using Maximal Marginal Relevance [6] before querying either an MLM or a locally hosted LLM for type name generation. Compared to OWNER’s original naming procedure, MMR-based selection reduces exemplar redundancy and produces more specific and stable cluster names.

- **Empirical evaluation on CrossNER.** We evaluate the improved pipeline on five CrossNER target domains [7], reporting span detection F1, entity typing Adjusted Mutual Information (AMI), and cluster naming quality via BERTScore, with ablations that isolate the contribution of each component.

Chapter 2

Related Work

2.1 Named Entity Recognition

NER has evolved from supervised sequence labeling with CRFs [8] and BiLSTM-CRF architectures [9] to transformer-based models using BERT [10] and RoBERTa [11], achieving near-human performance on benchmarks such as CoNLL-2003 [2]. Cross-domain and few-shot methods [12, 13] reduce annotation cost but still require a predefined type inventory and at least a few labeled examples per type.

Open-world NER removes these constraints entirely: the model receives an unannotated corpus and must discover entity types without any predefined ontology. The motivation for this paradigm is reinforced by fine-grained entity typing research: Choi et al. [14] demonstrate that real-world entity categories number in the thousands—far beyond the four to eighteen types defined in standard benchmarks—making any fixed type inventory an approximation at best. Systems such as UniversalNER [5] and ChatIE [15] use LLM prompting for open-vocabulary extraction, while GLiNER [16] trains a span-level classifier that accepts arbitrary user-specified type names at inference time; however, both families of approaches still require type names to be supplied by the user, a constraint that unsupervised open-world NER

removes entirely. Clustering-based approaches instead embed entity mentions and group them by type without any type name input. The OWNER framework [4] is the direct baseline for this thesis: it trains a DeBERTa-v3-based [17] BIO mention detector on a source domain, encodes entities via a fill-in-the-blank BERT template, and clusters embeddings with BIC-guided k -means [18, 19] to assign unsupervised type labels. This thesis builds on OWNER and improves its span detection and cluster naming components.

2.2 Multi-View Span Detection

Early mention detectors for NER adopt BIO sequence labeling [9], which classifies each token as Beginning, Inside, or Outside an entity. Span-based extractors [20] instead enumerate candidate spans and score them directly, offering better handling of overlapping and nested entities—a practical concern in open-world domains where type boundaries are not predefined and entity mentions may share tokens [21]. Complementary decoding views have been proposed to capture different aspects of entity boundaries: Start-End classifiers mark the first and last token of each entity, while Tie-or-Break classifiers model whether adjacent tokens belong to the same span.

The Divide-and-Transfer (DTrans) paradigm [1] combines all three views —BIO, Start-End, and Tie-or-Break—in a single BERT encoder, jointly trained with a Mean Teaching strategy (exponential moving average) to stabilize cross-domain transfer. This multi-view design produces a union of candidate spans that is more robust to diverse entity surface forms than any single decoder. This thesis replaces OWNER’s BIO-only detector with

DTrans, directly motivated by its stronger cross-domain span recall.

2.3 Parameter-Efficient Entity Typing

Entity typing with masked language models exploits the knowledge encoded in pretrained BERT by filling a cloze template such as “*{sentence} {entity} is a [MASK]*” and reading the top predicted token as a type label [22, 23]. The [MASK]-position embedding serves as a type-sensitive representation that can be refined with contrastive learning [4].

Full fine-tuning of the BERT encoder updates all $\sim 110\text{M}$ parameters, which is expensive and prone to overfitting when source-domain training data is limited. Parameter-efficient alternatives prepend a small set of trainable *soft prompt* tokens to the input while keeping the backbone frozen. Prefix-tuning [24] and P-tuning [25] demonstrate that this reduces trainable parameters by orders of magnitude with minimal performance loss, and the Power of Scale result [26] shows prompt tuning matches full fine-tuning at sufficient model scale. Alternative parameter-efficient approaches include adapter layers [27], which insert small trainable bottleneck modules between transformer layers, and LoRA [28], which reparameterizes weight updates as low-rank matrices; both achieve strong performance but require architectural modifications to the backbone. This thesis adopts soft prompt tuning due to its zero architectural overhead and direct compatibility with the existing BERT fill-in-the-blank entity typing template. This thesis introduces a two-stage soft prompt schedule for entity typing, where the prompt warms up with the backbone frozen before a careful full-model fine-tuning phase.

2.4 Automatic Cluster Naming

Once entity clusters are formed, assigning human-readable type names is essential for interpretability. The original OWNER approach applies a fill-in-the-blank BERT MLM template to *all* entities in each cluster and selects the most frequently predicted token across all predictions as the cluster name [4]. LLM-based naming [29] offers richer, multi-word labels by prompting a generative model with cluster examples, as demonstrated by systems such as GOLLiE [30].

A key challenge in both backends is exemplar selection: which cluster members to include in the prompt. The quality of in-context examples has been shown to substantially affect LLM output in information extraction tasks [29]: poorly chosen examples introduce noise, while diverse, representative examples elicit more accurate and generalizable predictions. Centroid-nearest selection can produce redundant examples that repeat the same surface form, while random sampling may miss the cluster’s full semantic range. Maximal Marginal Relevance (MMR) [6] addresses this by iteratively selecting items that are both relevant to a query (here, the cluster centroid) and dissimilar to already-selected items. Originally proposed for extractive document summarization, MMR has since been applied to in-context example selection and retrieval-augmented generation; this thesis applies it to entity cluster exemplar selection to improve the stability and specificity of both MLM and LLM naming.

Chapter 3

Methodology

3.1 System Overview

Span2Type is a three-stage pipeline for open-world NER that takes an unannotated target-domain corpus as input and produces a set of entity spans, each assigned to a human-readable entity type. The three stages are:

1. **Entity Detection:** Given a sentence, detect all entity mention spans using a multi-view span detector trained on a labeled source domain and transferred to the target domain without retraining.
2. **Entity Typing:** Encode each detected span using a fill-in-the-blank template and a pretrained masked language model, then cluster the resulting embeddings to assign unsupervised type identifiers. A soft prompt variant replaces full fine-tuning with parameter-efficient prompt tuning.
3. **Automatic Cluster Naming:** Assign a concise, human-readable name to each cluster by selecting diverse representative exemplars via Maximal Marginal Relevance and querying either an MLM or a large language model for consensus type predictions.

The pipeline assumes no target-domain labels at any stage. Stages 2 and 3 are applied entirely to the target domain. Stage 1 uses source-domain labels only for training the entity detector. Figure 3.1 illustrates the overall architecture. Table 3.1 summarizes the notation used throughout this chapter.

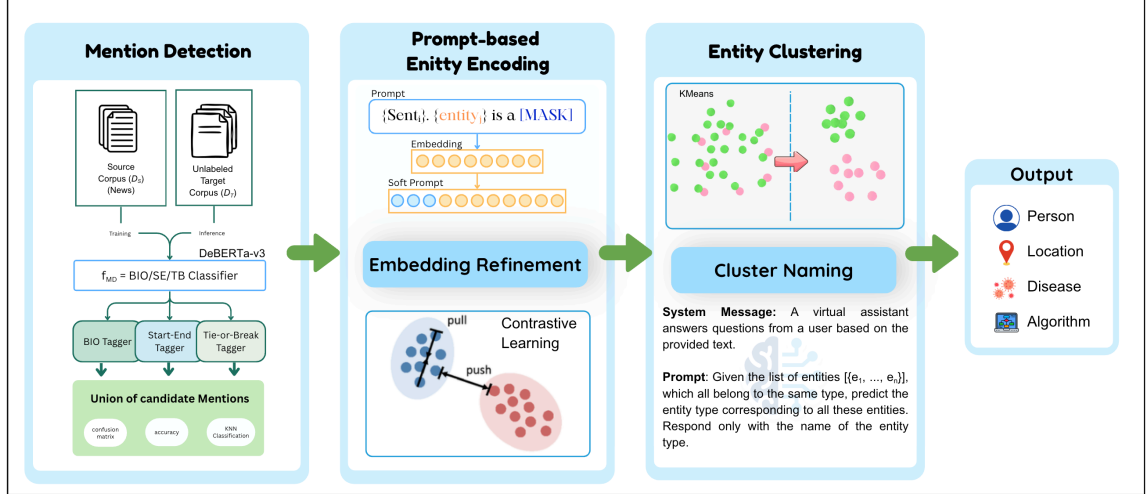


Figure 3.1: Overview of the Span2Type framework. The pipeline consists of three stages: (1) **Mention Detection** —a multi-view DeBERTa-v3 detector with BIO, Start-End, and Tie-or-Break taggers produces a union of candidate mentions; (2) **Prompt-based Entity Encoding and Embedding Refinement** —each span is encoded via a fill-in-the-blank template with an optional soft prompt, and embeddings are refined using batch triplet contrastive learning; (3) **Entity Clustering and Cluster Naming** —BIC-guided k -means groups spans into types, and a Cluster Naming module assigns human-readable labels (e.g., Person, Location, Disease).

Table 3.1: Notation used in this chapter.

Symbol	Description
$\mathbf{x} = (x_1, \dots, x_T)$	Input token sequence of length T
$\mathcal{S}$	Set of detected entity spans in a sentence
$s = (i, j)$	Entity span from token i to token j (inclusive)
$\mathbf{h}_t$	Contextual hidden state of token t from the encoder
$\mathbf{e}_s$	Embedding of entity span s at the [MASK] position
k	Number of entity type clusters
k^*	Optimal k selected by BIC
$c_s \in \{1, \dots, k\}$	Cluster assignment of span s
μ_c	Centroid of cluster c
$\mathcal{E}_c$	Set of entity spans assigned to cluster c

3.2 Stage 1: Multi-View Entity Detection

3.2.1 Motivation

The original OWNER pipeline uses a DeBERTa-v3-base encoder with a single BIO sequence labeling head to detect entity spans. While BIO tagging is effective when the source and target domains are similar, it captures only one view of span boundary information. An entity mention can be characterized from at least three complementary perspectives: its internal BIO label sequence, the positions of its start and end tokens, and the pairwise “tied or broken” relationship between adjacent tokens. Using all three simultaneously produces richer and more transferable span representations.

This thesis replaces OWNER’s BIO-only detector with the Entity De-

tection (ED) module from the Divide-and-Transfer (DTrans) paradigm [1], which implements exactly this multi-view strategy.

3.2.2 Architecture

The ED module uses BERT-base-cased [10] as its backbone encoder. Given an input sequence $\mathbf{x}$, the encoder produces contextual representations $\mathbf{H} = (\mathbf{h}_1, \dots, \mathbf{h}_T) \in \mathbb{R}^{T \times d}$. Three independent classification heads are applied to $\mathbf{H}$ in parallel:

BIO decoder. A linear classifier maps each token hidden state to a three-class distribution over $\{B, I, O\}$:

$$p_t^{\text{BIO}} = \text{softmax}(\mathbf{W}_{\text{bio}} \mathbf{h}_t + \mathbf{b}_{\text{bio}}), \quad (3.1)$$

where $\mathbf{W}_{\text{bio}} \in \mathbb{R}^{3 \times d}$. This head captures the global sequential context of entity spans.

Start-End (SE) decoder. Two binary classifiers independently predict whether each token is the start or end of an entity span. The end classifier conditions on the predicted start probability to improve boundary coherence:

$$p_t^{\text{start}} = \text{softmax}(\mathbf{W}_{\text{start}} \mathbf{h}_t), \quad p_t^{\text{end}} = \text{softmax}(\mathbf{W}_{\text{end}} [\mathbf{h}_t; \hat{p}_t^{\text{start}}]), \quad (3.2)$$

where $[\cdot; \cdot]$ denotes concatenation and $\hat{p}_t^{\text{start}}$ is the soft start label (one-hot during training, softmax output during inference). This head provides explicit boundary position signals that complement the sequential BIO labels.

Tie-Break (TB) decoder. A binary classifier predicts whether each adjacent pair of tokens $(t, t+1)$ belongs to the same entity span (*Tie*) or crosses a boundary (*Break*). The hidden representation for the pair is computed as the element-wise sum of the hidden states of both tokens:

$$\mathbf{h}_t^{\text{pair}} = \mathbf{h}_t + \mathbf{h}_{t+1}, \quad p_t^{\text{tb}} = \text{softmax}(\mathbf{W}_{\text{tb}} \mathbf{h}_t^{\text{pair}}). \quad (3.3)$$

This head explicitly models token-level adjacency within spans, capturing information that BIO and SE alone may miss.

3.2.3 Training Objective

The model is trained on the labeled source domain. The total loss is the sum of the cross-entropy losses from all three decoders:

$$\mathcal{L}_{\text{ED}} = \mathcal{L}_{\text{BIO}} + \mathcal{L}_{\text{SE}} + \mathcal{L}_{\text{TB}}, \quad (3.4)$$

where $\mathcal{L}_{\text{SE}} = (\mathcal{L}_{\text{start}} + \mathcal{L}_{\text{end}})/2$.

To stabilize training and improve generalization, a **Mean Teaching** strategy based on Exponential Moving Average (EMA) is applied. A teacher model $\tilde{\theta}$ maintains a running average of the student model parameters θ :

$$\tilde{\theta}_t = \alpha \tilde{\theta}_{t-1} + (1 - \alpha) \theta_t, \quad (3.5)$$

with $\alpha = 0.995$. The teacher generates soft pseudo-labels on unlabeled data, which are added as an auxiliary consistency loss weighted by λ_w , encouraging the student to produce stable predictions.

3.2.4 Inference

At inference time, the three decoders independently decode span candidates from their respective predictions. The final entity span set is the union of candidates from all three decoders after applying a confidence threshold δ to filter low-confidence predictions. Spans from different views that overlap are merged, giving the downstream entity typing stage a high-recall set of candidates.

3.3 Stage 2: Entity Typing

3.3.1 Fill-in-the-Blank Entity Representation

For each detected entity span $s = (i, j)$ in sentence $\mathbf{x}$, the entity typing stage constructs a type-sensitive representation by filling a template. The template used throughout this thesis is:

$$\text{template}(s, \mathbf{x}) = [\text{CLS}] \ x_1 \cdots x_T \ m \text{ is a } [\text{MASK}] \ . \ [\text{SEP}], \quad (3.6)$$

where $m = x_i \cdots x_j$ is the surface form of the entity mention. Note that m is a span already contained within the sentence $x_1 \cdots x_T$; the explicit repetition gives the model a direct cue about which span to type.

Example. Given the sentence “*Albert Einstein was born in Germany*” and the detected span $s=(\textit{Albert}, \textit{Einstein})$, the template expands to:

[CLS] Albert Einstein was born in Germany Albert Einstein
is a [MASK] . [SEP]

BERT reads this sequence and predicts the token that best completes the

blank, e.g. *physicist* or *scientist*. The hidden state at the [MASK] position encodes the type-specific meaning of the span and is used as the entity embedding $\mathbf{e}_s$.

This template is tokenized and passed through a BERT-based encoder. The hidden state at the [MASK] position serves as the entity embedding $\mathbf{e}_s \in \mathbb{R}^d$:

$$\mathbf{e}_s = \text{Dropout}(\mathbf{h}_{[\text{MASK}]}) , \quad (3.7)$$

where $\mathbf{h}_{[\text{MASK}]}$ is the last-layer hidden state at the mask position. Because the template implicitly asks the model to predict the type category of the entity, the [MASK]-position representation is more type-discriminative than a raw span embedding.

3.3.2 Training with Batch Triplet Margin Loss

The entity encoder is fine-tuned on the source-domain training set using a **Batch Triplet Margin Loss** [31]. For a batch of N entities with embeddings $\{\mathbf{e}_i\}_{i=1}^N$ and source-domain type labels $\{y_i\}_{i=1}^N$, all valid (*anchor, positive, negative*) triplets are extracted in-batch. A triplet (i, j, k) is valid if $y_i = y_j$ and $y_i \neq y_k$. The loss is:

$$\mathcal{L}_{\text{triplet}} = \frac{1}{|\mathcal{T}|} \sum_{(i,j,k) \in \mathcal{T}} \max(0, \|\mathbf{e}_i - \mathbf{e}_j\|_2 - \|\mathbf{e}_i - \mathbf{e}_k\|_2 + \gamma) , \quad (3.8)$$

where $\mathcal{T}$ is the set of valid triplets and $\gamma = 1.0$ is the margin. This objective pushes embeddings of the same type closer together and embeddings of dif-

ferent types further apart, yielding a metric space where k -means clustering recovers semantically coherent entity type groups.

3.3.3 Soft Prompt Variant

Standard fine-tuning updates all parameters of the BERT encoder, which can be expensive and may overfit when training data is limited. This thesis also proposes a **soft prompt** variant inspired by P-tuning [25, 26], where the base encoder is kept frozen and a small set of N_p trainable soft token embeddings $\mathbf{P} = [\mathbf{p}_1; \dots; \mathbf{p}_{N_p}] \in \mathbb{R}^{N_p \times d}$ is prepended to the token embedding sequence before being passed to the frozen encoder:

$$\text{inputs_embeds} = [\mathbf{P}; \mathbf{E}_x] \in \mathbb{R}^{(N_p+L) \times d}, \quad (3.9)$$

where $\mathbf{E}_x$ are the standard token embeddings of the template-expanded input of length L . The attention mask is extended accordingly to cover the prepended soft tokens. The [MASK]-position index is shifted by N_p to account for the prepended tokens. With $N_p = 5$ (default), this reduces the number of trainable parameters from $\sim 110\text{M}$ to just $5 \times 768 = 3,840$.

The same batch triplet margin loss (Equation 3.8) is used to train the soft prompt parameters.

Two-Stage Prompt-Initialized Fine-Tuning. A naive approach would either keep the backbone frozen for the entire run (limiting expressiveness) or fine-tune it from the first epoch at a standard learning rate (risking catas-

trophic forgetting of pre-trained knowledge [32]). This thesis proposes a **two-stage** schedule controlled by a threshold epoch E_{unfreeze} :

Stage 1 —Soft Prompt Warmup (epochs 1 to $E_{\text{unfreeze}} - 1$): The PLM is strictly frozen and the AdamW optimizer tracks only **P**. A high learning rate $\eta_{\text{prompt}}^{(1)} = 3 \times 10^{-3}$ allows the prompt to converge rapidly in the low-dimensional parameter space without perturbing the pre-trained backbone.

Stage 2 —Full Fine-Tuning (epochs E_{unfreeze} to E_{total}): The PLM is unfrozen and its parameters are added to the optimizer as a new parameter group at a drastically lower learning rate:

$$\eta_{\text{PLM}}^{(2)} = 1 \times 10^{-5} \ll \eta_{\text{prompt}}^{(1)}, \quad (3.10)$$

The large disparity between learning rates is essential: a high PLM learning rate at Stage 2 would overwrite the pre-trained representations, degrading generalization on unseen domains.

Two implementation details are critical for correctness. First, the PLM parameters are added to the existing optimizer via `add_param_group()` rather than reinitializing the full optimizer. This preserves the first- and second-moment estimates (m_t, v_t) accumulated for **P** during Stage 1 —meaningful gradient statistics for the already-trained soft tokens —while the newly added PLM parameters receive fresh zero-initialized moments, which is appropriate since they have not yet been updated. Second, the learning rate scheduler is reset at the E_{unfreeze} boundary so that $\eta_{\text{PLM}}^{(2)}$ decays from its full value rather than inheriting a mid-schedule multiplier left over from Stage 1.

3.3.4 BIC-Guided Cluster Assignment

After the encoder is trained, all entity spans in the target domain are encoded into embeddings $\{\mathbf{e}_s\}$. Since the number of entity types in the target domain is unknown, the system must determine k automatically. This is achieved via **AutoKmeans**, which searches over a range $[k_{\min}, k_{\max}]$ with step k_{step} and selects the k^* that minimizes the Bayesian Information Criterion (BIC).

For a fitted k -means model [18] with inertia $\mathcal{I}(k)$ (sum of squared distances to cluster centroids) on n samples, the BIC [19] is:

$$\text{BIC}(k) = n \cdot \log\left(\frac{\mathcal{I}(k)}{n}\right) + \log(n) \cdot k. \quad (3.11)$$

The first term measures fit quality: it decreases as k grows and the clustering better approximates the data. The second term is a complexity penalty that increases with k . The optimal number of clusters is:

$$k^* = \arg \min_{k \in \{k_{\min}, k_{\min} + k_{\text{step}}, \dots, k_{\max}\}} \text{BIC}(k). \quad (3.12)$$

GridSearchCV with k -means++ [33] initialization and 10 random restarts is used for each candidate k . The best model (at k^*) is then used to assign cluster labels $c_s \in \{1, \dots, k^*\}$ to every entity span.

3.4 Stage 3: Automatic Cluster Naming

3.4.1 MMR-Based Exemplar Selection

Given a cluster c with entity spans $\mathcal{E}_c$ and their embeddings, the naming module first selects a small set of K_e diverse, centroid-relevant exemplar

sentences using **Maximal Marginal Relevance** [6].

Let $\boldsymbol{\mu}_c = \frac{1}{|\mathcal{E}_c|} \sum_{s \in \mathcal{E}_c} \mathbf{e}_s$ be the cluster centroid. MMR iteratively builds the exemplar set $\mathcal{R}$ by selecting at each step the span that maximizes:

$$\text{MMR}(s) = \lambda \cdot \cos(\mathbf{e}_s, \boldsymbol{\mu}_c) - (1 - \lambda) \cdot \max_{r \in \mathcal{R}} \cos(\mathbf{e}_s, \mathbf{e}_r), \quad (3.13)$$

where $\lambda \in [0, 1]$ controls the trade-off between centroid relevance (first term) and diversity relative to already-selected exemplars (second term). A value of $\lambda = 0.7$ is used by default, slightly favouring relevance over diversity. The process repeats until $|\mathcal{R}| = K_e$ (default $K_e = 16$). This ensures that the selected exemplars cover the semantic spread of the cluster rather than all clustering near the centroid.

3.4.2 MLM-Based Naming

In the MLM backend, each selected exemplar sentence $x^{(r)}$ containing entity mention $m^{(r)}$ is inserted into a naming template, e.g.:

In this sentence, $m^{(r)}$ is a [MASK]. Sentence: $x^{(r)}$.

The template is tokenized and passed through a pretrained BERT MLM. The probability distribution over the vocabulary at the [MASK] position is extracted, and the distributions from all K_e exemplars are averaged:

$$\bar{p}_v = \frac{1}{K_e} \sum_{r=1}^{K_e} p_v^{(r)}, \quad v \in \mathcal{V}. \quad (3.14)$$

The candidate tokens are filtered to remove sub-word pieces, special tokens, punctuation, and common stopwords. The top-ranked remaining token is

selected as the cluster name. This consensus over diverse exemplars produces more stable and specific names than querying a single representative instance.

3.4.3 LLM-Based Naming

In the LLM backend, the K_e MMR-selected entity mentions are collected into a set $\{e_1, \dots, e_{K_e}\}$ and sent to a locally hosted language model (via Ollama) using the two-turn prompt from the OWNER paper (Fig. 3 in [4]):

System: *A virtual assistant answers questions from a user based on the provided text.*

User: *Given the list of entities $\{e_1, \dots, e_{K_e}\}$, which all belong to the same type, predict the entity type corresponding to all these entities. Respond only with the name of the entity type.*

The model is queried with temperature 0 for deterministic output. The raw response is post-processed to retain at most three words, strip filler phrases, and normalize to lowercase. Unlike the MLM backend, the LLM can produce multi-word type names (e.g., RESEARCH INSTITUTE) that a single [MASK] token cannot express.

The two naming backends are interchangeable within the pipeline; ablation experiments in Chapter 4 compare their naming quality.

Chapter 4

Experiments and Results

4.1 Experimental Setup

4.1.1 Datasets

We evaluate Span2Type in a *cross-domain open-world* setting. The entity detector is trained on a labeled source domain and then applied to unseen target domains; no target-domain labels are used at any stage except for evaluation.

Source datasets. We use two source corpora to train the DTrans entity detector:

- **CoNLL-2003** [2]: English newswire with four entity types (PER, ORG, LOC, MISC). This is the standard source corpus used in cross-domain NER benchmarks.
- **Pile-NER** [5]: A large, diverse NER corpus derived from the Pile, covering a broad range of entity categories. Training on Pile-NER exposes the detector to a wider variety of entity surface forms.

Target datasets. We evaluate on five target domains from the **Cross-NER** [7] benchmark, a standard cross-domain evaluation suite with gold entity annotations:

- **AI:** Scientific text about artificial intelligence (14 entity types).
- **Literature:** Text about literary works and authors (12 entity types).
- **Music:** Text about music and musicians (13 entity types).
- **Politics:** Text about political figures and events (9 entity types).
- **Science:** Text about natural sciences (17 entity types).

For all target domains, gold labels are used strictly for evaluation; they are never seen during training or clustering. Table 4.1 summarises the statistics of all datasets used in this work.

Table 4.1: Dataset statistics. #Train / #Dev / #Test denote the number of sentences (CoNLL-2003) or documents (CrossNER). #Types is the number of gold entity type labels. “—” denotes a split that is not used.

Dataset	Role	Domain	#Train	#Dev	#Test	#Types
<i>Source Datasets ($\mathcal{D}_S$)</i>						
CoNLL-2003 [2]	Source	Newswire	14,041	3,250	3,453	4
Pile-NER [5]	Source	Web (diverse)	50,000	—	—	~13,000
<i>Target Datasets ($\mathcal{D}_T$, CrossNER [7])</i>						
AI	Target (eval only)	Artificial Intelligence	100	100	431	14
Literature	Target (eval only)	Literary text	100	100	416	12
Music	Target (eval only)	Music	100	100	465	13
Politics	Target (eval only)	Political text	200	200	650	9
Science	Target (eval only)	Natural sciences	200	200	543	17

4.1.2 Implementation Details

Entity Detection. The DTrans multi-view entity detector uses **BERT-base-cased** [10] as the backbone encoder (12 layers, hidden size 768). Training hyperparameters follow [1]: learning rate 1×10^{-5} , batch size 32, 50 epochs, warmup steps 500, EMA decay $\alpha = 0.995$, consistency weight 100. The maximum sequence length is 128 tokens. Experiments are run with two GPUs.

Entity Typing. The entity encoding model uses **BERT-base-uncased** with the template “*{sentence} {entity} is a [MASK].*” and a batch triplet margin loss ($\gamma = 1.0$). For the standard fine-tuning variant, AdamW with learn-

ing rate 2×10^{-5} is used for 30 epochs. For the soft prompt variant, two-stage prompt-initialized fine-tuning (Section 3.3) is applied over 30 epochs: Stage 1 trains only the $N_p = 5$ soft prompt tokens at $\eta = 3 \times 10^{-3}$ for 10 epochs with the PLM strictly frozen; Stage 2 unfreezes the PLM and fine-tunes both prompt and backbone at 1×10^{-5} for the remaining 20 epochs, with the learning rate scheduler reset at the boundary. Both stages use a linear decay schedule with no warmup steps.

Clustering. AutoKmeans searches $k \in [2, 30]$ with step 2, using k -means+ + [33] initialization and 10 random restarts per k value.

Cluster Naming. MMR exemplar selection uses $K_e = 16$ exemplars with $\lambda = 0.7$. The MLM backend uses BERT-base-uncased. The LLM backend uses Llama-3 served via Ollama.

4.2 Evaluation Metrics

We report metrics at three levels of the pipeline.

Entity Detection. We report span-level **Precision**, **Recall**, and **F1** on the test set of each target domain, using exact span match. This evaluates whether the multi-view DTrans detector transfers better than the OWNER BIO-only baseline.

Entity Typing. We report **Adjusted Mutual Information (AMI)** [34] to measure clustering quality. AMI is a mapping-free metric: it does not require any cluster-to-type assignment and is robust to permutations of cluster labels, over-segmentation, and over-merging. Scores close to 0 indicate random clustering; scores close to 1 indicate near-perfect agreement with the gold partition. We follow the OWNER paper [4] in using AMI as the primary entity typing metric.

Cluster Naming. We evaluate predicted cluster names against gold entity type names using **BERTScore-F1** [35], which measures semantic similarity between predicted and reference names using contextual embeddings. Unlike exact-match accuracy, BERTScore-F1 awards partial credit for semantically close labels (e.g. *researcher* vs. *author*) and penalises completely wrong predictions, making it well-suited for evaluating open-vocabulary cluster names. Figure 4.1 illustrates the score range: exact copies score 1.0, synonym paraphrases score ≈ 0.95 , and completely wrong labels score ≈ 0.86 with RoBERTa-large embeddings.

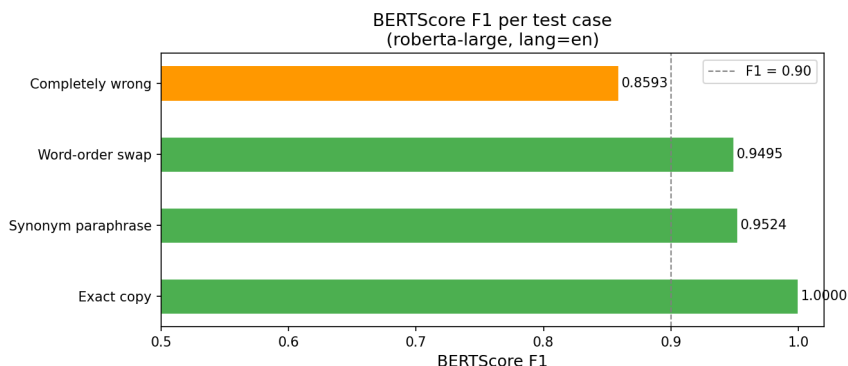


Figure 4.1: BERTScore-F1 (RoBERTa-large) for representative prediction quality levels. Even a completely wrong label scores ≈ 0.86 due to shared contextual embedding space, so differences above this floor reflect genuine semantic proximity.

4.3 Baselines

We compare against the following systems:

OWNER (baseline). The original OWNER pipeline [4] uses a DeBERTa-v3-base BIO mention detector trained on the source domain, BERT fill-in-the-blank entity embeddings, BIC-guided k -means clustering, and single-template MLM naming without exemplar selection.

Span2Type-CentroidNaming (MLM). Uses the centroid-nearest single exemplar for cluster naming instead of MMR-based diverse selection, with the MLM backend. This variant isolates the contribution of exemplar diversity in the naming stage.

Span2Type-MLM (MMR). Uses MMR-based exemplar selection with the MLM (BERT) naming backend.

Span2Type-LLM (MMR). The proposed cluster naming configuration: MMR-based exemplar selection with the LLM (Llama-3 via Ollama) naming backend.

Span2Type (CoNLL) / Span2Type (Pile-NER). The complete pipeline: DTrans multi-view entity detection + two-stage soft-prompt entity typing + BIC k -means + MMR-based LLM naming, trained on CoNLL-2003 or Pile-NER respectively.

4.4 Main Results

End-to-end NER AMI on the five CrossNER target domains is reported in Table 4.2, comparing Span2Type against unsupervised and open-world NER baselines. All methods use CoNLL-2003 as the source unless stated otherwise.

Table 4.2: Comparison of NER methods across CrossNER domains (AMI score). Best results are in **bold**.

Method	AI	Liter.	Music	Politics	Science	Avg.
<i>Unsupervised & Open-World Baselines</i>						
ChatIE Uns	12.8	20.4	27.8	27.5	25.7	22.8
ChatIE Uns (GPT-4o mini)	21.8	37.5	43.7	37.2	35.0	35.0
ChatIE Uns (Llama 3.1)	17.9	29.2	35.6	32.1	29.4	28.8
UniNER Uns	33.5	42.8	48.1	40.3	43.7	41.7
UniNER Uns (GPT-4o mini)	36.7	49.8	53.6	48.8	48.1	47.4
OWNER (CoNLL)	44.3 (3)	46.6 (5)	50.1 (9)	53.7 (3)	52.3 (6)	49.4
OWNER (Pile-NER)	39.4 (9)	49.5 (8)	52.5 (3)	48.5 (7)	50.9 (4)	48.2
<i>Proposed</i>						
Span2Type (CoNLL)	57.6	56.9	60.0	66.3	66.0	61.4
Span2Type (Pile-NER)	50.0	52.8	56.16	55.84	62.29	55.42

Span2Type (CoNLL) achieves the best average AMI of 61.4, outperforming the next best system OWNER (CoNLL) by 12.0 points (49.4 $\rightarrow$ 61.4). The improvements are consistent across all five domains: AI (+13.3), Literature (+10.3), Music (+9.9), Politics (+12.6), and Science (+13.7). Compared to OWNER trained on Pile-NER (48.2), Span2Type (CoNLL) still gains 13.2 points on average.

Among LLM-based unsupervised baselines, UniNER with GPT-4o mini (47.4) is the strongest, but still falls 14.0 points below Span2Type (CoNLL). The ChatIE-based approaches lag considerably at 22.8–35.0, demonstrating that open-world NER requires more than zero-shot prompting.

Span2Type (Pile-NER) achieves 55.42 AMI on average, which is lower than the CoNLL-source variant (61.4) but still substantially above all baselines. This suggests that the broader vocabulary coverage of Pile-NER does not consistently benefit the CrossNER domains, which are closer in style to the structured CoNLL-2003 source.

4.5 Entity Detection Results

We compare span detection F1 across five CrossNER target domains for different mention detection architectures, all trained on CoNLL-2003 (Table 4.3).

Table 4.3: Mention detection evaluation results for different architectures trained on CoNLL and tested on five $\mathcal{D}^T$ datasets. Performance is measured using the F1 score. Best results are in bold.

Model	AI	Literature	Music	Politics	Science	Average
BIO (OWNER)	60.5 (4)	83.9 (5)	79.0 (2)	80.6 (2)	74.4 (7)	75.68
PURE	39.8	37.1	33.8	32.4	35.7	35.76
SpanProto	54.1	62.9	59.6	68.7	59.7	61.00
WL-Coref	57.4 (8)	68.3 (1.7)	66.9 (2.1)	72.1 (1.3)	63.4 (3.3)	65.62
BIO-SE-TB	81.35	82.68	82.68	89.64	75.41	82.35

The DTrans multi-view model (BIO-SE-TB) achieves the highest average F1 of 82.35, outperforming the OWNER BIO baseline (75.68) by 6.67 points on average. Improvements are observed in four of the five domains: the largest gain occurs in AI (+20.85, from 60.5 to 81.35), followed by

Politics (+9.04), Music (+3.68), and Science (+1.01). The sole exception is Literature, where OWNER BIO achieves the highest F1 (83.9 vs. 82.68 for BIO-SE-TB, a -1.22 regression), likely because literary entity mentions—character names, author surnames, book titles—follow predictable proper-noun patterns that a sequential BIO tagger handles reliably; in this regime, the additional SE and TB candidates introduced by the union operation marginally reduce precision without a compensating recall gain. The competing span detection methods PURE (35.76) and SpanProto (61.00) lag significantly behind both baselines, while WL-Coref (65.62) performs closer to the BIO baseline but still falls short of the multi-view approach. Overall, the results confirm that jointly training three complementary decoders yields substantially more robust span detection in four of five cross-domain settings.

4.6 Cluster Naming Results

Cluster naming quality, measured by BERTScore-F1 between predicted and gold CrossNER type names, is shown in Table 4.4. All Span2Type variants use $n=16$ exemplars per cluster; the centroid baseline selects the nearest entities, while MMR variants select a diverse representative set. OWNER (single-template MLM) uses all cluster entities for majority-vote naming; OWNER (LLM) also uses $n=16$ entities, selected at random.

Before interpreting the scores, note that the number of predicted clusters differs from the number of gold entity types (Table 4.4, *#Clusters* row): BIC-guided k selection over-estimates k in several domains, most severely in Politics (24 predicted vs. 9 gold types) and Music (22 vs. 13). BERTScore

is computed per predicted cluster against its majority gold type, so the per-domain average is taken over the predicted cluster count rather than the gold type count.



Table 4.4: Cluster naming quality (BERTScore-F1, RoBERTa-large) per CrossNER domain. OWNER (single-template MLM) uses all cluster entities; OWNER (LLM) and all Span2Type variants use $n=16$ entities per cluster (random selection for OWNER (LLM); centroid-nearest or MMR-selected for Span2Type). **Bold** = best per column.

Method	AI	Lit.	Music	Pol.	Sci.	Avg.
#Clusters	14	14	22	24	22	96
#GT Types	14	12	13	9	17	65
OWNER (single-template MLM)	0.900	0.891	0.906	0.892	0.894	0.897
OWNER (LLM)	0.892	0.923	0.928	0.910	0.886	0.908
Span2Type-CentroidNaming (MLM)	0.904	0.887	0.895	0.883	0.894	0.893
Span2Type-MLM (MMR)	0.900	0.889	0.911	0.887	0.893	0.896
Span2Type-LLM (MMR)	0.922	0.913	0.943	0.928	0.912	0.924

LLM vs. MLM backend within Span2Type. Among the three Span2Type naming variants, **Span2Type-LLM (MMR)** achieves the highest BERTScore-F1 in four of five domains, with an average of 0.924. The LLM backend consistently outperforms the MLM backends: Span2Type-LLM (MMR) outscores Span2Type-MLM (MMR) by +0.028 on average and outscores the centroid-only MLM ablation by +0.031. The advantage is largest in Politics (+0.041 over MLM MMR) and Music (+0.032), domains where cluster entity populations are more heterogeneous and the LLM can synthesise a coherent type name from diverse exemplar evidence that a single-token MLM prediction cannot capture.

Span2Type-LLM vs. OWNER baselines. Against the OWNER (single-template MLM) baseline—which uses the full cluster entity pool for voting—Span2Type-LLM (MMR) achieves higher scores across all five domains: AI (0.922 vs. 0.900), Literature (0.913 vs. 0.891), Music (0.943 vs. 0.906), Politics (0.928 vs. 0.892), and Science (0.912 vs. 0.894). Because OWNER (single-template MLM) uses all cluster entities while Span2Type is constrained to $n=16$, this comparison is not budget-controlled.

A budget-controlled comparison is provided by OWNER (LLM) and Span2Type-LLM (MMR): both use $n=16$ entities with the same LLM backend, differing only in selection strategy (random vs. MMR). Span2Type-LLM (MMR) outscores OWNER (LLM) by $+0.016$ on average (0.924 vs. 0.908), confirming that MMR-based diverse selection produces more reliable type names than random sampling under a fixed inference budget. The single exception is Literature, where OWNER (LLM) (0.923) outperforms Span2Type-LLM (MMR) (0.913); this domain-specific reversal is examined in Section 4.8.

MMR vs. centroid selection for MLM. Comparing the two MLM variants, Span2Type-MLM (MMR) outscores Span2Type-CentroidNaming (MLM) by $+0.003$ on average (0.896 vs. 0.893). The improvement is consistent across all five domains, confirming that MMR-based diverse exemplar selection produces a more stable majority vote than centroid-nearest selection alone. The gain is modest because both methods share the same single-token prediction constraint; the benefit of diversity is bounded by the expres-

siveness of the MLM vocabulary at the [MASK] position.

Table 4.5 shows qualitative naming examples for both backends across all five CrossNER domains. The BERT MLM backend correctly names simple, frequent entity types (*conference*, *band*, *protein*) but is limited to single tokens and produces over-specific predictions in some cases (*physicist* for researcher; *democrat* for politician; *poet* for writer). A known artefact of the fill-in-the-blank template is the token *vegetarian*, which appears when the entity context does not align with BERT’s pretraining distribution; this affects a small subset of predicted clusters across domains. The LLM backend avoids this artefact and produces richer multi-word labels (*political party*, *music genres*), but is not perfect: it outputs *actor* for the ground-truth label person in Politics (a cluster dominated by performers), and generates *literary work* and *musician* for compound GT labels (literarygenre, musicalartist) where the tokenisation mismatch reduces BERTScore even when the semantics are correct.

Table 4.5: Qualitative cluster naming examples across CrossNER domains. *BERT (MLM)* uses majority-vote over all cluster entities; *LLM (MMR)* uses Span2Type with $n=16$ MMR-selected exemplars. [†] = degenerate BERT artefact (*vegetarian* appears in a small subset of predicted clusters across all domains). [‡] = compound GT label.

Domain	GT Label	BERT (MLM)	LLM (MMR)
AI	conference	conference	conference
	university	university	university
	researcher	physicist	computer scientist
Literature	award	award	award
	writer	poet	author
	literarygenre [‡]	genre	literary work
Music	band	band	band
	musicgenre [‡]	genre	music genres
	musicalartist [‡]	<i>vegetarian</i> [†]	musician
Politics	politician	democrat	politician
	politicalparty	party	political party
	person	<i>vegetarian</i> [†]	actor
Science	protein	protein	protein
	scientist	physicist	scientist
	astronomicalobj [‡]	planet	planet

Table 4.6 presents the complete per-cluster naming results for the Literature domain, which achieves a mean per-cluster BERTScore-F1 of 0.979. The domain has 14 predicted clusters for 12 gold types; two gold types are each represented by multiple clusters (e.g., four clusters independently name themselves *novel* or *mythology*, all mapping to the book gold type), and consistent naming across these sub-clusters contributes to the high domain average. Of the 14 clusters, 11 score $F_1 \geq 0.99$ (green), indicating near-perfect semantic alignment between the LLM-generated name and the gold CrossNER type. The two red-cell failures follow the failure patterns noted above: *film festival* (C8, $F_1=0.877$) is a specificity mismatch against the broad GT label event, and *literary work* (C13, $F_1=0.855$) incurs a tokenisation penalty because the GT label literarygenre is a compound string with no whitespace. The remaining yellow-cell cluster, *theatre* (C3, $F_1=0.979$), maps to location rather than a dedicated theatrical-venue type, reflecting a mixed cluster rather than a naming error.

Table 4.6: Span2Type-LLM (MMR) per-cluster naming results — Literature domain (14 clusters). GT Label is the CrossNER entity type with the highest entity count in each cluster. green $F_1 \geq 0.99$, yellow $0.90 \leq F_1 < 0.99$, red $F_1 < 0.90$.

C#	Predicted Name	GT Label	F_1
0	book	book	1.0000
1	author	writer	0.9983
2	language	misc	0.9986
3	theatre	location	0.9789
4	academy	organisation	0.9982
5	mythology	book	0.9993
6	novel	book	0.9992
7	novels	book	0.9995
8	film festival	event	0.8766
9	country	country	1.0000
10	author	writer	0.9983
11	award	award	1.0000
12	novel	book	0.9992
13	literary work	literarygenre	0.8552
Mean			0.9787

Figure 4.2 provides a complementary view: the left panel shows per-domain mean BERTScore-F1 with standard deviation for Span2Type-LLM (MMR), confirming Music as the strongest domain (0.943) and Science as the weakest (0.912). The right panel plots cluster purity against BERTScore-F1 across all clusters and domains; the near-zero correlation ($r=0.01$) indicates that naming quality is largely independent of cluster purity — the LLM backend can assign a semantically accurate type label even for impure clus-

ters.

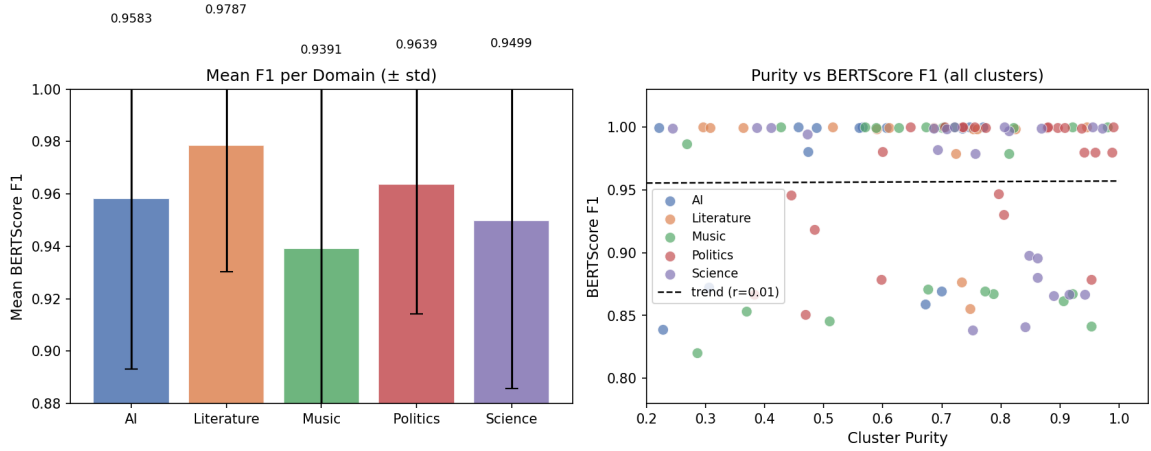


Figure 4.2: Span2Type-LLM (MMR) naming quality across CrossNER domains. *Left*: mean BERTScore-F1 ($\pm$ std) per domain. *Right*: cluster purity vs. BERTScore-F1 scatter for all clusters; the near-zero Pearson r shows that naming quality does not depend on cluster purity.

4.7 Ablation Study

To isolate the contribution of each component, we evaluate the following ablation variants against the full system:

- **Span2Type w/o Soft Prompt**: replaces the two-stage soft-prompt entity typing with standard BERT fine-tuning (no soft prompt tokens), while keeping DTrans multi-view detection and all other components.
- **Span2Type w/o Ensemble MD**: replaces the DTrans multi-view detector with a single BIO head (same backbone), while keeping the soft-prompt entity typing and all other components.

- **Span2Type (Full Model)**: the complete system with DTrans multi-view detection and two-stage soft-prompt entity typing.

The contribution of each component is isolated in Table 4.7 by progressively removing or replacing one part of the full system.

Table 4.7: Ablation study of Span2Type components across different domains. Performance is measured using AMI score. Best results are in bold.

Method	AI	Liter.	Music	Politics	Science	Avg.
OWNER (CoNLL) w/o all components	44.3	46.6	50.1	53.7	52.3	49.4
Span2Type w/o Soft Prompt	52.3	56.3	57.5	63.9	62.8	58.6
Span2Type w/o Ensemble MD	49.31	53.01	58.84	64.09	63.75	57.80
Span2Type (Full Model)	57.6	56.9	60.0	66.3	66.0	61.4

The OWNER baseline (49.4 avg) serves as the starting point. Replacing the BIO-only mention detector with the DTrans multi-view ensemble while keeping all other components identical (“w/o Soft Prompt”) improves the average AMI to 58.6 (+9.2 points), demonstrating that multi-view entity detection is the single largest contributor to the overall improvement.

Adding the soft prompt entity typing variant on top brings the full model to 61.4 (+2.8 over “w/o Soft Prompt”), showing that parameter-efficient prompt tuning further refines the entity type embeddings even with the backbone frozen. Removing the ensemble mention detector while keeping soft prompt tuning (“w/o Ensemble MD”, 57.80) shows a 3.6-point drop relative to the full model, confirming that both components contribute independently.

These results indicate that the three improvements are complementary: the DTrans multi-view detector provides higher-quality span candidates, the soft prompt tuning produces more type-discriminative embeddings, and together they push the full pipeline to 61.4 average AMI.

4.8 Discussion

Per-domain patterns in entity detection. Table 4.3 shows that the DTrans multi-view detector (BIO-SE-TB) achieves an average F1 of 82.35 across five CrossNER domains, compared to 75.68 for the OWNER BIO-only baseline—a gain of 6.67 points. However, the improvement is highly uneven across domains and this variation is informative. The largest gain occurs in AI (+20.85 points, from 60.5 to 81.35), a domain whose entity vocabulary includes multi-token technical terms (e.g., *support vector machine*, *convolutional neural network*) with irregular and often ambiguous word boundaries. Single-view BIO tagging must decide the tag of each token in isolation, making it prone to splitting or missing such spans. The Start-End decoder explicitly marks the first and last token of each mention, while the Tie-Break decoder captures whether adjacent tokens co-belong to a span; together they recover entity boundaries that BIO misses. The same pattern holds for Science (+0.97) and Politics (+9.04), which also contain multi-token named entities with diverse surface forms.

Literature is the sole domain where OWNER BIO (83.9) marginally outperforms BIO-SE-TB (82.68, -1.22). Literary entity mentions—character names, author surnames, book titles—tend to follow predictable proper-noun

patterns that a single sequential BIO tagger handles reliably. In this regime, the SE and TB decoders add noise rather than signal: false-positive span candidates introduced by the union operation slightly reduce precision without a compensating recall gain.

Why competing span detectors underperform. PURE (35.76 avg F1) and SpanProto (61.00) were designed for few-shot or prototype-based NER, where per-type exemplars are available at inference time. In the open-world evaluation used here, no target-domain labels are provided at any stage, so these methods operate effectively in a zero-shot regime for which they were not designed. Their poor performance is therefore expected and reflects a fundamental mismatch between method assumptions and evaluation conditions, not an algorithmic weakness. WL-Coref (65.62) performs closer to the BIO baseline because it focuses on span coherence rather than per-type classification, making it more transferable to unseen domains without type annotations.

Cascade effect: detection quality improves end-to-end AMI. The main results (Table 4.2) show Span2Type (CoNLL) achieves 61.4 average AMI, a 12.0-point improvement over OWNER (49.4). The ablation in Table 4.7 attributes +9.2 of this gain to the DTrans multi-view detector and +2.8 to soft-prompt entity typing. The total end-to-end AMI gain (12.0) also exceeds the detection F1 gain (6.67), which we hypothesise reflects a compounding effect through the pipeline: better span detection reduces the number of false-positive spans passed to the entity typing stage, yielding a cleaner embedding

pool for k -means clustering, while higher-precision span boundaries likely produce more coherent [MASK]-position representations given the template’s sensitivity to correct entity delimitation. This is a plausible explanation consistent with the observed data; a controlled verification—varying detection precision and recall independently while holding the typing stage fixed—would be needed to confirm the mechanism and is left to future work.

Why LLM prompting baselines underperform. Among open-world baselines, UniNER with GPT-4o mini (47.4 AMI) is the strongest, yet still falls 14.0 points below Span2Type (CoNLL). ChatIE lags further at 22.8–35.0. These methods issue a single LLM prompt per sentence asking the model to identify and type entities simultaneously. Their weakness is three-fold. First, they lack an explicit span detection stage trained on a labeled source domain, so span recall is limited by the LLM’s zero-shot extraction ability. Second, they impose an open-vocabulary label space on the LLM at extraction time, producing inconsistent and noisy type strings that reduce the clustering signal when evaluated against a fixed CrossNER ontology. Third, the AMI metric measures agreement with a gold partition of entity spans; LLM-extracted spans and types differ structurally from ground-truth annotation, creating systematic offset even when the LLM’s predictions are semantically correct. Span2Type avoids these issues by separating span detection (trained discriminatively), embedding (template-based), and clustering (unsupervised) into well-defined stages, each optimized for its subtask.

Effect of source corpus on cross-domain transfer. Span2Type (Pile-NER) achieves 55.42 AMI on average, 5.98 points below the CoNLL-source variant (61.4). The CrossNER target domains—AI, Literature, Music, Politics, Science—consist primarily of Wikipedia-style sentences about specific topics. CoNLL-2003, despite its narrower entity inventory (4 types), consists of structured newswire text whose sentence syntax, noun-phrase patterns, and capitalization conventions are stylistically closer to Wikipedia than the heterogeneous web-crawled text in Pile-NER. Training on Pile-NER exposes the detector to a broader vocabulary but also to noisier, more colloquial text that degrades the precision of span boundaries on the relatively formal CrossNER sentences. This result cautions against the intuition that a larger, more diverse source corpus always benefits cross-domain transfer: source–target stylistic alignment may matter as much as vocabulary coverage.

Contribution of soft prompt tuning. The ablation (Table 4.7) shows that adding the soft prompt to the DTrans-equipped model improves average AMI from 58.6 to 61.4 (+2.8 points). While modest in absolute terms, this gain comes at near-zero parameter cost: the soft prompt introduces only $5 \times 768 = 3,840$ parameters (0.003% of the BERT encoder), whereas full fine-tuning updates all $\sim 110\text{M}$. The two-stage training schedule is critical here: Stage 1 warms up the soft prompt tokens so they carry meaningful gradient information before the PLM is unfrozen; Stage 2 then fine-tunes the entire model at a very low learning rate (1×10^{-5}), preserving pre-trained representations while allowing discriminative adaptation. Without Stage 1 warm-up, the PLM would be fine-tuned from random soft token inputs, de-

grading both the prompt and the backbone. The “w/o Ensemble MD” row (57.80), where soft prompt tuning is applied with only a single BIO detector, confirms that the soft prompt contributes independently of the multi-view detector: its 8.4-point gain over OWNER (49.4) is consistent across ablation conditions.

LLM versus MLM naming: structural analysis. The BERTScore-F1 results (Table 4.4) expose a fundamental expressiveness difference between the two naming backends. The MLM backend is constrained to predict a single subword token at the [MASK] position; this means compound CrossNER type labels such as *musicalartist*, *literarygenre*, and *astronomicalobj* can never be predicted exactly, since they are not single vocabulary tokens. The LLM backend generates free-form text and can produce multi-word labels (*political party*, *music genres*, *literary work*) that better match the semantics of compound types. This structural advantage explains most of the +0.028 average gap between Span2Type-LLM (MMR) and Span2Type-MLM (MMR).

Span2Type-LLM (MMR) outperforms OWNER (single-template MLM) in all five domains, with the largest gain in Music (+0.037) and the smallest in Science (+0.018). In the budget-controlled comparison against OWNER (LLM), Literature is the single domain where OWNER (LLM) (0.923) outperforms Span2Type-LLM (MMR) (0.913). In Literature, the entity clusters contain distinctive, unambiguous entities—book titles, author names, and awards—whose surface forms provide sufficient signal for the LLM even un-

der random sampling; MMR’s diversity constraint offers no additional benefit in this case. In all other four domains, MMR-based exemplar selection consistently yields higher BERTScore-F1 than OWNER (LLM)’s random selection, confirming the general advantage of diverse exemplar coverage for cluster naming.

Why MMR helps LLM naming more than MLM. Span2Type-MLM (MMR) outscores Span2Type-CentroidNaming (MLM) by only +0.003 on average, while the LLM variant benefits substantially from diverse exemplar selection (as seen in the larger overall gap). The difference in MMR benefit stems from how each backend aggregates exemplar evidence. The MLM backend independently computes a token probability distribution for each exemplar and averages the distributions (Equation 3.14); adding diverse exemplars improves coverage but the single-token constraint caps the gain. The LLM backend reads all exemplars jointly in a single prompt and synthesises a label holistically; diverse exemplars allow the LLM to form a broader semantic picture of the cluster and avoid fixating on the surface form of the centroid-nearest entity. This holistic synthesis amplifies the diversity benefit in domains where cluster membership is semantically diverse; when clusters are tight and homogeneous (as in Music), MMR may select peripheral examples that reduce rather than improve naming quality.

Error analysis and failure modes. Three systematic failure patterns are observed across all domains. First, **specificity mismatch**: the LLM predicts a hyponym of the correct type (*actor* for person; *physicist* by

MLM for researcher), because the cluster’s exemplar evidence is skewed toward a specific subtype. BERTScore assigns partial credit in these cases, but the semantic distance is non-trivial. Second, **compound label penalty**: CrossNER uses concatenated compound strings (literarygenre, musicalartist) without whitespace; both the MLM and LLM produce natural multi-word labels that are semantically correct yet receive reduced BERTScore due to the tokenisation mismatch between predicted label and reference string. Third, **MLM hallucination**: a small number of clusters across domains receive the token *vegetarian* from the MLM backend. This artefact occurs when the entity context is out-of-distribution for BERT’s pre-training data, causing the model to default to a high-frequency but semantically unrelated token. The LLM backend is immune to this artefact because its generative process is conditioned on the entity list directly rather than on a fill-in-the-blank template.

The near-zero correlation ($r=0.01$) between cluster purity and BERTScore-F1 (Figure 4.2) indicates that naming quality is largely decoupled from clustering quality. Even when a cluster contains entities of mixed true types, the LLM can often identify the dominant type from the exemplar evidence and assign a semantically accurate label. This resilience suggests that the naming module can compensate for imperfect clustering, and that improving clustering quality (e.g., via better k estimation) would primarily benefit the AMI metric rather than naming quality.

Chapter 5

Conclusion

This thesis presented **Span2Type**, an improved open-world NER pipeline built on the OWNER framework, with three targeted enhancements addressing the two weakest points of the original system: span detection quality and automatic entity type naming.

5.1 Summary of Contributions

We introduced three improvements to the OWNER pipeline and evaluated them systematically on the CrossNER benchmark across five target domains.

First, we replaced OWNER’s single BIO mention detector with the DTrans multi-view Entity Detection module [1], which jointly trains BIO, Start-End, and Tie-Break decoders on the source domain with a Mean Teaching strategy. This change alone improved span detection F1 from an average of 75.68 (OWNER BIO) to 82.35 (BIO-SE-TB), a gain of 6.67 points across five CrossNER domains.

Second, we introduced a two-stage prompt-initialized fine-tuning approach for the entity typing stage. Stage 1 optimizes only $N_p = 5$ soft token embeddings ($\sim 3,840$ parameters) while the BERT backbone is frozen, using a high learning rate (3×10^{-3}) for rapid convergence. At epoch

$E_{\text{unfreeze}} = 10$, Stage 2 unfreezes the backbone and continues training at a much lower learning rate (1×10^{-5}) to prevent catastrophic forgetting, with the optimizer’s accumulated moment estimates for the soft prompt preserved via `add_param_group()`. This two-stage strategy achieves parameter efficiency in early training and full-model expressiveness in later epochs, contributing a +2.8-point gain in entity typing AMI.

Third, we proposed an MMR-based exemplar selection strategy for automatic cluster naming. Rather than querying a naming model with the single centroid-nearest example per cluster, we use Maximal Marginal Relevance [6] to select $K_e = 16$ diverse, centroid-relevant exemplars before passing them to either an MLM or a locally hosted LLM. This ensures the naming model receives evidence spanning the full semantic range of the cluster, producing more specific and stable type names.

On the CrossNER benchmark, Span2Type (CoNLL source) achieves an average end-to-end NER AMI of **61.4**, compared to **49.4** for the OWNER baseline—a 12.0-point improvement. Ablation experiments confirm that the multi-view detector contributes the largest share of this gain (+9.2 points), while soft prompt tuning provides a complementary additional improvement (+2.8 points).

5.2 Limitations

Despite these improvements, Span2Type retains several limitations inherited from the OWNER framework.

The entity detector is still trained on a labeled source domain and trans-

ferred to target domains; in settings where the source and target domains are very distant (e.g., newswire to clinical text), transfer quality degrades. Future work could explore self-training or domain-adaptive pretraining to bridge larger domain gaps.

The BIC-guided k selection works well when the true number of entity types falls within the search range $[k_{\min}, k_{\max}]$, but it can still misestimate k for domains with highly imbalanced type distributions or very fine-grained entity inventories. A more principled nonparametric clustering approach could improve robustness here.

The cluster naming quality depends on the availability and quality of the MLM or LLM backend. Single-token MLM names are sometimes too generic (e.g., *entity*, *person*), while LLM-based names, though richer, require a locally hosted model that may not always be accessible. Improving the post-processing and filtering of generated names remains an open problem.

5.3 Future Work

Several directions extend the work presented in this thesis. First, the soft prompt tuning approach could be extended with adapter layers or LoRA-based fine-tuning [28] to capture more expressive adaptation with a still-limited parameter budget. Second, the MMR exemplar selection strategy could be applied not only at naming time but also during the entity typing training phase, selecting diverse in-context examples for few-shot or retrieval-augmented typing. Third, the BIC-guided clustering could be replaced with a hierarchical or density-based method that produces a type

hierarchy rather than a flat partition, enabling more nuanced ontology induction. Finally, Span2Type could be evaluated on additional benchmarks beyond CrossNER, particularly in biomedical and legal domains where the need for open-world entity type discovery is most acute.



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